

Name :- Ramani Vemula
Div :- B(BI)
Roll No. :- 9
PRN :- 202201040011

OS Assignment - 1

discuss

Date 30/09/24

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Q.1.

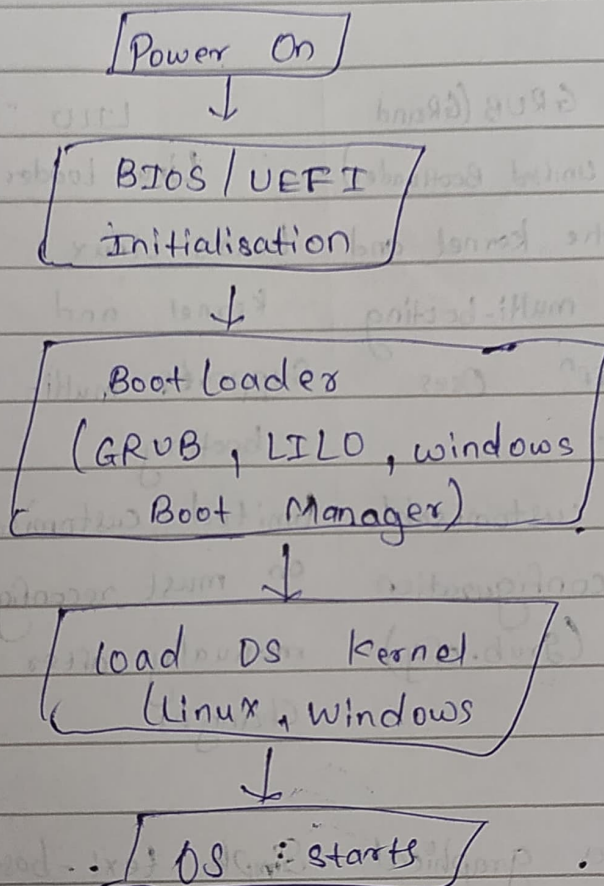
→ Similarities :-

- ① Process Management :- All operating systems manage processes, ensuring that CPU resources are allocated efficiently and processes are executed in an orderly manner.
- ② Includes handling process creation, scheduling and termination.

③ Similarities :- Windows and Linux

Aspect	Windows	Linux
① Core functions	Resource management, multitasking, Security, user Interface [GUI/CLI]	Resource management, multitasking, Security, user Interface.
② User Interaction	offers both GUI & CLI (command prompt, Powershell)	offers both GUI (GNOME, KDE) and CLI (Bash)
③ Multitasking	Supports running multiple applications simultaneously	Supports running multiple app ⁿ simultaneously
④ Security Management	Provide user authentication, encryption and access control	Provide user authentication, encryption & access control
⑤ Networking	Supports TCP/IP, DNS, DHCP and other networking products.	Fully Supports TCP/IP, DNS, DHCP and other essential networking products

* Diagram :-



Q.2.

Aspect	GRUB (Grand Unified Bootloader)	Lilo Linux Loader	Windows Boot Manager
① Fun.			
① function	Loads the kernel and supports multi-booting bet ⁿ diff ⁿ Oses	Loads Linux kernel and supports multi-booting	Loads the windows OS or other windows-compatible Oses.
② customisation	Highly customisable with configuration files (grub.cfg)	limited customisation must reconfigure manually after changes	Minimal customisation Primarily managed through windows settings.
③ UI	Provides a graphical menu and command line interface for selecting OS/kernel	Simple text-based interface for OS selection	Simple GUI for selecting between windows installations no command line interface.
④ Support for multiple Oses	Excellent support for multiple OS and diff ⁿ file systems	Supports multiple Oses, but with limited flexibility compared to GRUB	Limited to window Oses and compatible versions (e.g. Linux via third-party tools).
⑤ Configuration complexity	Can be complex for some beginners due to extensive configuration options	Easier to configure but less flexible; requires installation after changes.	Easy to use with automatic configuration but lacks flexibility.
⑥ Boot time	slightly slower boot times due to additional features and flexibility	faster boot time due to simplicity and fewer features	Generally fast & optimised for windows booting

Difference :- Window & Linux

Aspect	Window	Linux
① Kernel Architecture	Hybrid Kernel combining monolithic & microkernel elements	Monolithic kernel with all core functions running in kernel space.
② File System	Primarily uses NTFS (New Tech file System)	Commonly uses ext4.
③ Software Management	Relies on executables ('.exe', '.msi') for software installation	user package managers (apt, yum) to install & update.
④ User Interface	Consistent GUI with Start Menu, Taskbar & File explorer	offers a variety of customizable desktop environments (GNOME, KDE).
⑤ Security	Implements Security via ACLS, uses authentication & encryption, Supports Secure Boot.	Strong emphasis on security with user Permissions, SELinux and various encryption process/options.